

MATH 411: Honors Complex Variables

Midterm Exam 1

Fall 2026 | Friday, October 9, 2026

Instructor: Manish Patnaik

Time: 3:00–3:50 PM (50 minutes) **Total: 25 points**

CAB 369

Student name: _____

Student ID: _____

Directions

1. This is a closed-book exam. No notes, textbooks, calculators, or electronic devices may be used.
2. Answer all four questions. Check that your exam has seven pages: this cover; Question 1 on page 2; Question 2 on pages 3–4; Question 3 on page 5; and Question 4 on pages 6–7.
3. Write clearly in the space provided. Show your work and justify your answers. Partial credit may be awarded for meaningful progress.
4. You may use results proved in class. State clearly any result you use and check its hypotheses.
5. If you continue an answer elsewhere in the exam, indicate where the continuation appears and label it with the question and part number.

For instructor use only

Question	Points available	Score
Question 1 (page 2)	7	
Question 2 (pages 3–4)	6	
Question 3 (page 5)	4	
Question 4 (pages 6–7)	8	
Total	25	

- 1. (7 points)** Let $U \subseteq \mathbb{C}$ be an (open) disc, and let $f = u + iv$ be holomorphic on U . Suppose that

$$u(x, y)^2 + v(x, y)^2 = 2u(x, y) \quad \text{for every } x + iy \in U.$$

Prove that f is constant on U .

2. (6 points)

- (a) **(2 points)** Determine the radius of convergence of

$$\sum_{n=1}^{\infty} n^2 z^{n^2}.$$

Justify your answer.

- (b) **(4 points)** Let $f(T) = \sum_{n \geq 0} a_n T^n$ and $h(T) = \sum_{n \geq 1} b_n T^n$ be convergent power series, with $h(0) = 0$. Assume that f has radius of convergence $r > 0$ and that $s > 0$ satisfies $\sum_{n \geq 1} |b_n| s^n \leq r$. Let $g = f \circ h$ be the composite power series, i.e.,

$$g(T) = \sum_{n \geq 0} a_n \left(\sum_{k=1}^{\infty} b_k T^k \right)^n.$$

- i. Show that g is a well-defined power series.

- ii. Show that g has radius of convergence at least s .

3. (4 points) Let $U \subseteq \mathbb{C}$ be an (open) disc, and let f and g be analytic on U . Suppose that

$$f(z)g(z) = 0 \quad \text{for every } z \in U.$$

Prove that either $f \equiv 0$ on U or $g \equiv 0$ on U .

4. (8 points) Justify your answers to each part.

(a) (3 points) Compute $\int_C \bar{z} dz$, where C is the counterclockwise-oriented unit circle.

- (b) **(5 points)** Let $R \subseteq \mathbb{C}$ be a closed rectangle, and let a be a point in its interior. Suppose f is continuous on an open neighborhood U of R and holomorphic on $U \setminus \{a\}$. Prove that

$$\int_{\partial R} f(z) dz = 0.$$

Hint: Use Cauchy–Goursat for rectangles and the ML estimate; isolate the point a in a small sub-rectangle.